Large totient fibers from smooth shifted primes: a conditional proof of Erdős's conjecture

Abstract

Let $g(n)$ be the number of positive integers m with $\varphi(m) = n$. We assume a hypothesis on smooth shifted primes: in $(2x, 5x]$ there are at least $x^{1-\eta}$ primes p with $p-1$ being x^{δ}-smooth. Under this hypothesis we show that for every $\varepsilon > 0$ there are infinitely many n with $g(n) > n^{1-\varepsilon}$. The argument is the classical product-and-pigeonhole method (Pomerance, Theorem B). We take the products of $k \approx x^{2\delta}$ such primes, and their totients form a family of smooth numbers that is small compared with $\binom{M}{k}$, so some value n is hit very often. The proof is split into elementary counting lemmas, explicit real-variable estimates, and a final assembly. Every lemma and the main theorem are machine-verified, with the smooth-shifted-primes hypothesis taken as an explicit assumption. The quantitative bound $g(n) \ll_{\eta} n^{1+\eta}$ of the source is not part of the formalization; only finiteness of the fibers is proved.

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1 Setting

Throughout, φ is Euler's totient function on the nonnegative integers, with $\varphi(0) = 0$. All sets of integers below are subsets of $\mathbb{N} = \{0, 1, 2, \dots\}$. For $p \geq 1$ the symbol $p-1$ has its usual meaning. For a real number t we write

$$\lfloor t \rfloor_+ = \max(\lfloor t \rfloor, 0) \in \mathbb{N}$$

for the floor truncated at 0. For $N \in \mathbb{N}$ we let $\ell_2(N)$ be the largest $e \in \mathbb{N}$ with $2^e \le N$ when $N \ge 1$, and we put $\ell_2(0) = 0$. Thus $\ell_2(N) = \lfloor \log_2 N \rfloor$ for $N \ge 1$. Powers a^s with a real exponent denote Mathlib's real power function, which is defined for every real base a and agrees with $\exp(s \log a)$ for $a > 0$. In most statements below the base is positive. The exceptions are Hypothesis (H), Lemma 2.15 and Lemma 2.17, where the base may be any real number; they are discussed where they occur.

Definition 1.1 (Fiber count)**.** For $n \in \mathbb{N}$ let

$$g(n) = \#\{m \in \mathbb{N} : \ m > 0, \ \varphi(m) = n\}.$$

The convention is that $g(n) = 0$ if this set were infinite. By Lemma 2.1 the set is always finite, so $g(n)$ is its genuine cardinality.

Definition 1.2 (Smoothness)**.** Let $m \in \mathbb{N}$ and $y \in \mathbb{R}$. We say m is *y-smooth* if every prime factor q of m satisfies $q \le y$. For $m > 1$ this means $P^+(m) \le y$. The integers 0 and 1 have no prime factors and are y-smooth for every y.

Definition 1.3 (Hypothesis (H))**.** *Hypothesis (H)* is the statement that for all real δ with $0 < \delta < 1/4$ and all real $\eta > 0$ there is a real x_0 such that for every real $x \ge x_0$

$$\#\{p : \ p \text{ prime}, \ 2x < p \le 5x, \ p - 1 \text{ is } x^\delta\text{-smooth}\} \ \ge \ x^{1-\eta}.$$

The statement is unchanged if x_0 is replaced by $\max(x_0, 1)$. Hence only $x \ge 1$ matters, and there x^δ and $x^{1-\eta}$ are the usual real powers. The counted set is finite because its elements are at most $5x$.

The problem states the result conditionally on (H), and Theorem 1.4 is exactly this conditional statement. In the source paper, (H) is Theorem 1.2, proved separately; it is not formalized here, so the unconditional conjecture would further require a proof of (H) as defined above.

Theorem 1.4. *Assume Hypothesis (H). Then for every real $\varepsilon > 0$ the set*

$$\{n \in \mathbb{N} : \ n > 0, \ g(n) > n^{1-\varepsilon}\}$$

is infinite.

This is the conclusion of Erdős's conjecture under (H), as the problem states it. The source also proves, as Lemma 1, that every fiber is finite and that $g(n) \ll_\eta n^{1+\eta}$. Only the finiteness is formalized here (Lemma 2.1); the upper bound $g(n) \ll_\eta n^{1+\eta}$ is neither proved nor used.

2 Lemmas

2.1 Fibers of the totient and products of primes

Lemma 2.1 (Finiteness of totient fibers)**.** *For every $n \in \mathbb{N}$ the set $\{m \in \mathbb{N} : m > 0, \ \varphi(m) = n\}$ is finite.*

Proof. *Case $n = 0$.* If $m > 0$ then $\varphi(m) \ge 1$, so the set is empty.

Case $n > 0$. Put $B = (n+1)^{(n+2)n}$. We show that every $m > 0$ with $\varphi(m) = n$ satisfies $m \le B$; this gives finiteness.

Write $m = \prod_{p \mid m} p^{a_p}$ with $a_p = v_p(m) \ge 1$. Fix a prime $p \mid m$ and put $a = a_p$. Since $p^a \mid m$, we have $\varphi(p^a) \mid \varphi(m) = n$; here we use the standard fact that $d \mid m$ implies $\varphi(d) \mid \varphi(m)$. Moreover $\varphi(p^a) = p^{a-1}(p-1)$. This gives two consequences:

- From $p-1 \mid n$ and $n>0$ we get $p-1 \le n$, that is, $p \le n+1$.
- From $p^{a-1} \mid n$ we get $p^{a-1} \le n$. By induction $j+1 \le 2^j$ for all $j \in \mathbb{N}$. Hence

$$a \le 2^{a-1} \le p^{a-1} \le n.$$

All prime factors of m lie in $\{0, 1, \dots, n+1\}$, so there are at most $n+2$ of them. Therefore $\sum_{p|m} a_p \le (n+2)n$, and

$$m = \prod_{p|m} p^{a_p} \le \prod_{p|m} (n+1)^{a_p} = (n+1)^{\sum_p a_p} \le (n+1)^{(n+2)n} = B. \qquad \square$$

Lemma 2.2 (Fiber count dominates any subset)**.** *Let $n \in \mathbb{N}$, and let $T \subset \mathbb{N}$ be a finite set such that every $m \in T$ satisfies $m > 0$ and $\varphi(m) = n$. Then $|T| \le g(n)$.*

Proof. By Lemma 2.1 the fiber $\{m > 0 : \varphi(m) = n\}$ is finite, so $g(n)$ is its cardinality. The set T is a subset of this fiber, hence $|T| \le g(n)$. $\square$

Lemma 2.3 (Totient of a squarefree product)**.** *Let Q be a finite set of primes. Then*

$$\varphi\Big(\prod_{p \in Q} p\Big) = \prod_{p \in Q} (p-1).$$

Proof. We use induction on $|Q|$. The empty case reads $\varphi(1) = 1$.

Suppose $Q = Q' \cup \{p\}$ with $p \notin Q'$. Each $q \in Q'$ is a prime different from p, so q is coprime to p. Hence $\prod_{q \in Q'} q$ is coprime to p. By multiplicativity of φ, the induction hypothesis and $\varphi(p) = p-1$,

$$\varphi\Big(p \prod_{q \in Q'} q\Big) = \varphi(p)\, \varphi\Big(\prod_{q \in Q'} q\Big) = (p-1) \prod_{q \in Q'} (q-1). \qquad \square$$

Lemma 2.4 (Products of distinct primes determine the set)**.** *Let Q_1, Q_2 be finite sets of primes with $\prod_{p \in Q_1} p = \prod_{p \in Q_2} p$. Then $Q_1 = Q_2$.*

Proof. For a finite set Q of primes, the set of prime factors of $\prod_{p \in Q} p$ is exactly Q. Hence

$$Q_1 = \{\text{prime factors of } \textstyle\prod_{Q_1} p\} = \{\text{prime factors of } \textstyle\prod_{Q_2} p\} = Q_2. \qquad \square$$

Lemma 2.5 (Pigeonhole for totients of k-fold products)**.** *Let S be a finite set of primes and let $k \in \mathbb{N}$ with $k \le |S|$. Let $V \subset \mathbb{N}$ be a finite set such that*

$$\prod_{p \in Q} (p-1) \in V \quad \text{for every } Q \subseteq S \text{ with } |Q| = k.$$

Then there is $Q \subseteq S$ with $|Q| = k$ and

$$\binom{|S|}{k} \;\le\; |V| \cdot g\Big(\prod_{p \in Q} (p-1)\Big).$$

Proof. Let $\mathcal{T}$ be the family of k-element subsets of S, and put $f(Q) = \prod_{p \in Q}(p-1)$. Then $|\mathcal{T}| = \binom{|S|}{k}$, and this number is positive because $k \le |S|$. By hypothesis f maps $\mathcal{T}$ into V, so

$$|\mathcal{T}| = \sum_{v \in V} |\mathcal{T}_v|, \qquad \mathcal{T}_v = \{Q \in \mathcal{T} : f(Q) = v\}.$$

If V were empty, $\mathcal{T}$ would be empty, contradicting $|\mathcal{T}| > 0$.

Choose $n \in V$ with $|\mathcal{T}_n|$ maximal. Then $|\mathcal{T}| \le |V|\,|\mathcal{T}_n|$. Since $|\mathcal{T}| > 0$, the family $\mathcal{T}_n$ is nonempty; pick $Q \in \mathcal{T}_n$. Then $Q \subseteq S$, $|Q| = k$ and $f(Q) = n$. It remains to show $|\mathcal{T}_n| \le g(n)$.

Consider the map $R \mapsto \prod_{p \in R} p$ on $\mathcal{T}_n$.

- It is injective by Lemma 2.4.
- Each image $m = \prod_{p \in R} p$ is positive.
- By Lemma 2.3, $\varphi(m) = \prod_{p \in R}(p-1) = n$.

Thus the image is a set of $|\mathcal{T}_n|$ positive integers in the fiber of n. By Lemma 2.2, $|\mathcal{T}_n| \le g(n)$. Hence

$$\binom{|S|}{k} \le |V|\, g(n) = |V|\, g(f(Q)). \qquad \square$$

2.2 Counting smooth values

Lemma 2.6 (Few smooth numbers up to N)**.** *Let $N \in \mathbb{N}$ and $y \in \mathbb{R}$. There is a finite set $V \subset \mathbb{N}$ with*

$$|V| \le (\ell_2(N) + 1)^{\lfloor y \rfloor_+}$$

such that every y-smooth integer n with $0 < n \le N$ belongs to V.

Proof. Let P be the set of primes in $\{1, \dots, \lfloor y \rfloor_+\}$, and put $L = \ell_2(N)$. Define

$$V = \Big\{ \prod_{p \in P} p^{e_p} : \ e \in \{0, \dots, L\}^P \Big\}.$$

Then

$$|V| \le (L+1)^{|P|} \le (L+1)^{\lfloor y \rfloor_+},$$

because $|P| \le \lfloor y \rfloor_+$ and $L + 1 \ge 1$.

Now let $0 < n \le N$ be y-smooth.

Prime factors of n lie in P. Every prime factor q of n is an integer with $q \le y$, hence $q \le \lfloor y \rfloor_+$, and so $q \in P$. Consequently

$$n = \prod_{p \in P} p^{v_p(n)},$$

since primes of P not dividing n contribute $p^0 = 1$.

Exponents are at most L. For $p \in P$ we have $p^{v_p(n)} \mid n$ and $p \ge 2$. Hence $2^{v_p(n)} \le p^{v_p(n)} \le n \le N$, which gives $v_p(n) \le L$.

So n is the value at the exponent vector $(v_p(n))_{p \in P} \in \{0, \dots, L\}^P$, and $n \in V$. $\square$

Lemma 2.7 (Products of smooth shifted primes)**.** *Let $Q \subset \mathbb{N}$ be finite, let $y \in \mathbb{R}$ and let $B \in \mathbb{N}$. Suppose that every $p \in Q$ satisfies $p \ge 2$ (that is, $p - 1 > 0$), $p - 1 \le B$, and that $p - 1$ is y-smooth. Then $n = \prod_{p \in Q}(p-1)$ satisfies the following:*

- *$n > 0$;*
- *$n \le B^{|Q|}$;*
- *n is y-smooth.*

Proof. Positivity and the bound $n \le \prod_{p \in Q} B = B^{|Q|}$ are immediate from the factorwise properties.

For smoothness, let q be a prime factor of n. Since q is prime and divides the product, q divides $p - 1$ for some $p \in Q$. As $p - 1 \ne 0$, q is a prime factor of $p - 1$, and $p - 1$ is y-smooth. Hence $q \le y$. $\square$

2.3 Elementary real estimates

Lemma 2.8 (Lower bound for binomial coefficients)**.** *For $M, k \in \mathbb{N}$ with $k \le M$,*

$$\Big(\frac{M}{k}\Big)^k \le \binom{M}{k}.$$

For $k = 0$, with the convention $M/0 = 0$, this reads $1 \le 1$.

Proof. Let $k \ge 1$. Since $k \le M$, we have $M/k \ge 1$.

Factorwise bound. For $0 \le j < k$,

$$M - j \;\ge\; M - \frac{M}{k}\, j = \frac{M}{k}(k - j),$$

because $(M/k)j \ge j$.

Product. All factors are nonnegative, so

$$k!\binom{M}{k} = \prod_{j=0}^{k-1}(M - j) \;\ge\; \prod_{j=0}^{k-1}\frac{M}{k}(k - j) = \Big(\frac{M}{k}\Big)^k k!.$$

Dividing by $k! > 0$ gives the claim. □

Lemma 2.9 (Logarithmic bound for the value count)**.** *Let $\delta > 0$ and $x \ge 1$ be real. Then*

$$\log\Big(\frac{x^{2\delta}\log(5x)}{\log 2} + 1\Big) \;\le\; \log 11 + (2\delta + 1)\log x.$$

Proof. Two numerical facts. First, $\log 5 < 2$, because

$$e^2 \ge \sum_{m=0}^{7}\frac{2^m}{m!} = \frac{155}{21} > 5.$$

Second, $\log 2 > 3/10$, because the exponential series with its standard tail bound gives

$$e^{3/10} \le \sum_{m=0}^{7}\frac{(3/10)^m}{m!} + \frac{(3/10)^8 \cdot 9}{8! \cdot 8} < 2.$$

In particular $10\log 2 > 3$.

Bound for $\log(5x)/\log 2$. Using $\log x \le x - 1 \le x$ and $x \ge 1$,

$$\log(5x) = \log 5 + \log x \le 2 + x \le 3x < 10\log 2 \cdot x.$$

Hence $\log(5x)/\log 2 \le 10x$.

Conclusion. Write

$$\frac{x^{2\delta}\log(5x)}{\log 2} + 1 = x^{2\delta}\Big(\frac{\log(5x)}{\log 2} + x^{-2\delta}\Big).$$

Both factors are positive; note $\log(5x) > 0$. Since $x^{-2\delta} \le 1$, the second factor is at most $10x + 1 \le 11x$. Taking logarithms,

$$\log\Big(\frac{x^{2\delta}\log(5x)}{\log 2} + 1\Big) \le 2\delta\log x + \log 11 + \log x.$$ □

Lemma 2.10 ($\log x$ and x^δ are eventually large)**.** *Let $\delta > 0$ and $a, b \in \mathbb{R}$. There is $x_0 \geq 1$ such that $a \leq \log x$ and $b \leq x^\delta$ for all $x \geq x_0$.*

Proof. Both $\log x$ and x^δ tend to $+\infty$ as $x \to \infty$. Choose thresholds for a and b separately, and take x_0 to be the maximum of these two thresholds and 1. □

Lemma 2.11 (Growth comparison)**.** *Let $\delta > 0$, $c \in \mathbb{R}$ and $d > 0$. There is $x_0 \geq 1$ such that for all $x \geq x_0$*

$$x^\delta \log\Big(\frac{x^{2\delta}\log(5x)}{\log 2} + 1\Big) + c\,x^{2\delta} \;<\; d\,x^{2\delta}\log x.$$

Proof. Apply Lemma 2.10 with

$$a = \max(1,\, 2(|c|+1)/d), \qquad b = 2(\log 11 + 2\delta + 1)/d.$$

Let $x \geq x_0$, and put $u = x^\delta > 0$ and $L = \log x$. Then

$$L \geq 1, \qquad dL \geq 2(|c|+1), \qquad du \geq 2(\log 11 + 2\delta + 1).$$

Key inequality. Since $L \geq 1$ and $\log 11 > 0$,

$$\log 11 + (2\delta+1)L \leq (\log 11 + 2\delta + 1)L \leq \tfrac{1}{2}duL.$$

Also $cu < (|c|+1)u \leq \frac{1}{2}dLu$. Adding,

$$\log 11 + (2\delta+1)L + cu < duL.$$

Conclusion. Multiply by $u > 0$ and use $x^{2\delta} = u^2$:

$$u\big(\log 11 + (2\delta+1)L\big) + c\,x^{2\delta} < d\,x^{2\delta}L.$$

By Lemma 2.9, $u\log(x^{2\delta}\log(5x)/\log 2 + 1) \leq u\big(\log 11 + (2\delta+1)L\big)$, and the claim follows. □

Lemma 2.12 (Pointwise passage to $k = \lfloor x^{2\delta}\rfloor_+$)**.** *Let δ, ε, x be real with $\delta > 0$, $4\delta < \varepsilon < 1$ and $x > 1$. Assume*

$$x^\delta \log\Big(\frac{x^{2\delta}\log(5x)}{\log 2} + 1\Big) + (1-\varepsilon)\log 5 \cdot x^{2\delta} < \frac{\delta}{2}\,x^{2\delta}\log x. \tag{$*$}$$

Put $k = \lfloor x^{2\delta}\rfloor_+$. Then

- $1 \leq k$;
- $k \leq x^{1-\delta}$;
- *and*

$$x^\delta \log\Big(\frac{k\log(5x)}{\log 2} + 1\Big) + (1-\varepsilon)k\log(5x) \;<\; k\log\Big(\frac{x^{1-\delta}}{k}\Big).$$

Proof. Put $y = x^{2\delta}$ and $L = \log x > 0$.

The first two claims. We have $y \geq 1$, so $k \geq 1$, $k \leq y$ and $y < k+1$. Since $3\delta < \frac{3}{4}\varepsilon < 1$, we have $2\delta \leq 1-\delta$, so $k \leq y \leq x^{1-\delta}$.

Lower bound for the right side. From $k \leq y$ we get $\log k \leq 2\delta L$. Hence

$$k\log(x^{1-\delta}/k) = k\big((1-\delta)L - \log k\big) \geq (1-3\delta)kL.$$

Upper bound for the left side. The first term is at most $x^\delta \log(y \log(5x)/\log 2+1)$, by monotonicity of log and $k \le y$. The second term equals

$$(1-\varepsilon)\log 5 \cdot k + (1-\varepsilon)kL \le (1-\varepsilon)\log 5 \cdot y + (1-\varepsilon)kL.$$

By $(*)$ and $y < k+1$, the left side is therefore less than

$$\tfrac{\delta}{2}(k+1)L + (1-\varepsilon)kL.$$

Comparison. It suffices to show $\frac{\delta}{2}(k+1) \le (\varepsilon - 3\delta)k$. This holds because

$$(\varepsilon-3\delta)k = (\varepsilon-4\delta)k + \delta k \ge \delta k \ge \tfrac{\delta}{2}(k+1),$$

where we used $k \ge 1$. □

Lemma 2.13 (Eventual inequality for $k = \lfloor x^{2\delta} \rfloor_+$)**.** *Let $\delta > 0$ and $4\delta < \varepsilon < 1$. There is $x_0 \ge 1$ such that for every real $x \ge x_0$ the three conclusions of Lemma 2.12 hold with $k = \lfloor x^{2\delta} \rfloor_+$.*

Proof. Take x_0' from Lemma 2.11 with $c = (1-\varepsilon)\log 5$ and $d = \delta/2$, and put $x_0 = x_0' + 1$. For $x \ge x_0$ we have $x > 1$ and $(*)$ holds. Now apply Lemma 2.12. □

Lemma 2.14 (Logarithm of the value bound)**.** *Let $x, \delta, \varepsilon \in \mathbb{R}$ with $x \ge 1$ and $\varepsilon < 1$, and let $k \in \mathbb{N}$. Put $N = \lfloor 5x \rfloor_+$, $A = \ell_2(N^k) + 1$ and $t = \lfloor x^\delta \rfloor_+$. Then $A^t\,(N^k)^{1-\varepsilon} > 0$, and*

$$\log\Big(A^t\,(N^k)^{1-\varepsilon}\Big) \;\le\; x^\delta \log\Big(\frac{k\log(5x)}{\log 2} + 1\Big) + (1-\varepsilon)k\log(5x).$$

Proof. We have $1 \le N \le 5x$, so both factors are positive. Moreover

$$\log(A^t(N^k)^{1-\varepsilon}) = t\log A + (1-\varepsilon)k\log N.$$

First term. From $2^{\ell_2(N^k)} \le N^k$ we get

$$\ell_2(N^k) \le \frac{k\log N}{\log 2} \le \frac{k\log(5x)}{\log 2},$$

hence $\log A \le \log(k\log(5x)/\log 2 + 1)$. This right side is nonnegative, and $t \le x^\delta$, so $t\log A \le x^\delta \log(k\log(5x)/\log 2 + 1)$.

Second term. Since $1-\varepsilon > 0$ and $\log N \le \log(5x)$, we have $(1-\varepsilon)k\log N \le (1-\varepsilon)k\log(5x)$. □

Lemma 2.15 (Logarithm of the binomial lower bound)**.** *Let $x, \delta \in \mathbb{R}$ and $k, M \in \mathbb{N}$ with $k \ge 1$, $k \le x^{1-\delta}$ and $x^{1-\delta} \le M$. Then*

$$k\log\Big(\frac{x^{1-\delta}}{k}\Big) \le \log\binom{M}{k}.$$

No sign condition on x is imposed. The lemma holds for every real x, with $x^{1-\delta}$ read as Mathlib's real power function. The proof uses only the hypotheses $k \le x^{1-\delta}$ and $k > 0$, not the value of x.

Proof. We have $k \le M$, so $\binom{M}{k} > 0$. Also $0 < x^{1-\delta}/k \le M/k$. Hence, using Lemma 2.8 in the last step,

$$k\log\frac{x^{1-\delta}}{k} \le k\log\frac{M}{k} = \log\Big(\frac{M}{k}\Big)^k \le \log\binom{M}{k}.$$

□

Lemma 2.16 (Main asymptotic inequality)**.** *Let $\delta > 0$ and $4\delta < \varepsilon < 1$. There is $x_0 \geq 1$ with the following property. Let $x \geq x_0$ be real and let $M \in \mathbb{N}$ with $x^{1-\delta} \leq M$. Put $k = \lfloor x^{2\delta} \rfloor_+$ and $N = \lfloor 5x \rfloor_+$. Then*

- *$1 \leq k \leq M$;*
- *and*

$$(\ell_2(N^k) + 1)^{\lfloor x^\delta \rfloor_+} (N^k)^{1-\varepsilon} < \binom{M}{k}.$$

Proof. Take x_0 from Lemma 2.13. Then $1 \leq k \leq x^{1-\delta} \leq M$.

By Lemma 2.14, the logarithm of the left side is at most

$$x^\delta \log\bigl(k \log(5x)/\log 2 + 1\bigr) + (1-\varepsilon) k \log(5x).$$

By Lemma 2.13, this quantity is strictly smaller than $k \log(x^{1-\delta}/k)$. By Lemma 2.15, that in turn is at most $\log \binom{M}{k}$.

Both sides are positive, so exponentiating gives the claim. □

2.4 Assembly

Lemma 2.17 (A finite set of smooth shifted primes)**.** *Assume (H). Let $0 < \delta < 1/4$ and $\eta > 0$. There is x_0 such that for every real $x \geq x_0$ there is a finite set $S \subset \mathbb{N}$ with $|S| \geq x^{1-\eta}$ such that every $p \in S$ satisfies:*

- *p is prime;*
- *$2x < p \leq 5x$;*
- *$p - 1$ is x^δ-smooth.*

The statement does not require $x_0 \geq 1$. If $x_0 < 1$, the powers x^δ and $x^{1-\eta}$ for $x < 0$ are Mathlib's real power at a negative base. The threshold may, however, always be taken to be at least 1, and the proof below does so.

Proof. Apply (H) with δ and $\eta/2$ to obtain x_0', and put $x_0 = \max(x_0', 1)$. Let $x \geq x_0$.

The set S of primes p with $2x < p \leq 5x$ and $p - 1$ being x^δ-smooth lies in $\{0, \ldots, \lfloor 5x \rfloor_+\}$, so it is finite. By (H),

$$|S| \geq x^{1-\eta/2} \geq x^{1-\eta},$$

where the last inequality uses $x \geq 1$. □

Lemma 2.18 (A large fiber above x)**.** *Let $S \subset \mathbb{N}$ be finite, let $x, \delta, \varepsilon \in \mathbb{R}$ with $x \geq 1$ and $\varepsilon < 1$, and let $k \in \mathbb{N}$. Assume that every $p \in S$ is prime with $2x < p \leq 5x$ and $p - 1$ being x^δ-smooth, that $1 \leq k \leq |S|$, and that, with $N = \lfloor 5x \rfloor_+$,*

$$(\ell_2(N^k) + 1)^{\lfloor x^\delta \rfloor_+} (N^k)^{1-\varepsilon} < \binom{|S|}{k}. \tag{$**$}$$

Then there is $n \in \mathbb{N}$ with $n > x$, $n > 0$ and $g(n) > n^{1-\varepsilon}$.

Proof. Setting up the value set. Each $p \in S$ satisfies $p \geq 2$, and $p \leq N$ because p is an integer with $p \leq 5x$. Hence $0 < p - 1 \leq N$.

Let V be the set given by Lemma 2.6 for N^k and $y = x^\delta$. Then $|V| \leq (\ell_2(N^k) + 1)^{\lfloor x^\delta \rfloor_+}$. For $Q \subseteq S$ with $|Q| = k$, Lemma 2.7 with $B = N$ shows that $\prod_{p \in Q}(p-1)$ is positive, at most N^k and x^δ-smooth. Hence this product lies in V.

Choosing n. By Lemma 2.5 there is $Q \subseteq S$ with $|Q| = k$ and

$$\binom{|S|}{k} \leq |V|\, g(n), \qquad n = \prod_{p \in Q}(p-1).$$

By Lemma 2.7, $0 < n \leq N^k$.

The bound $n > x$. Since $k \geq 1$, there is some $p_0 \in Q$. Then $p_0 - 1 \mid n$, so

$$n \geq p_0 - 1 > 2x - 1 \geq x.$$

The bound $g(n) > n^{1-\varepsilon}$. Suppose instead $g(n) \leq n^{1-\varepsilon}$. Since $1 - \varepsilon > 0$ and $n \leq N^k$,

$$|V|\, g(n) \leq |V|\, n^{1-\varepsilon} \leq |V|\, (N^k)^{1-\varepsilon} \leq (\ell_2(N^k) + 1)^{\lfloor x^\delta \rfloor_+} (N^k)^{1-\varepsilon} < \binom{|S|}{k} \leq |V|\, g(n),$$

where the strict inequality is $(**)$. This is a contradiction. □

Lemma 2.19 (Arbitrarily large good n)**.** *Assume (H). Let $0 < \varepsilon < 1$ and $C \in \mathbb{R}$. There is $n \in \mathbb{N}$ with $n > C$, $n > 0$ and $g(n) > n^{1-\varepsilon}$.*

Proof. Put $\delta = \varepsilon/8$. Then $0 < \delta < 1/4$ and $4\delta = \varepsilon/2 < \varepsilon$.

Take x_0 from Lemma 2.17 with this δ and $\eta = \varepsilon/8$. Take x_1 from Lemma 2.16 with δ and ε. Let $x = \max(x_0, x_1, C, 1)$.

By Lemma 2.17 there is a finite set S of primes in $(2x, 5x]$ with $p - 1$ being x^δ-smooth and $|S| \geq x^{1-\delta}$. By Lemma 2.16 with $M = |S|$, the integer $k = \lfloor x^{2\delta} \rfloor_+$ satisfies $1 \leq k \leq |S|$ together with $(**)$.

Lemma 2.18 now gives $n > x \geq C$ with $n > 0$ and $g(n) > n^{1-\varepsilon}$. □

Lemma 2.20 (The case $\varepsilon < 1$)**.** *Assume (H) and let $0 < \varepsilon < 1$. Then the set $\{n \in \mathbb{N} : n > 0,\ g(n) > n^{1-\varepsilon}\}$ is infinite.*

Proof. Suppose the set were finite. Then it would be bounded above by some $M \in \mathbb{N}$. Lemma 2.19 with $C = \max(M, 0)$ gives an element $n > M$ of the set, which is a contradiction. □

3 Proof of the theorem

Proof of Theorem 1.4. Let $\varepsilon > 0$.

Case $\varepsilon < 1$. This is Lemma 2.20.

Case $\varepsilon \geq 1$. By Lemma 2.20, the set

$$\{n > 0 : g(n) > n^{1/2}\}$$

is infinite. For every integer $n \geq 1$ we have $n^{1-\varepsilon} \leq n^{1/2}$, because $1 - \varepsilon \leq 1/2$. Hence this infinite set is contained in $\{n > 0 : g(n) > n^{1-\varepsilon}\}$, which is therefore infinite. □

Relation to the source proof. The argument is the product-and-pigeonhole proof of the source. Products of $k \approx x^{2\delta}$ smooth shifted primes have smooth totients, and these totients form a set of size $x^{o(k)}$. In contrast there are $\binom{M}{k} \geq (M/k)^k$ products, so some totient value is attained very often. The formalization deviates in the following points.

- *Normalization of the parameters.* The source normalizes by primes $p \leq X$, with more than $X^{1-\delta}$ of them. The formalization works directly with the primes in $(2x, 5x]$ supplied by (H), at least $x^{1-\delta}$ of them. It uses $k = \lfloor x^{2\delta} \rfloor_+$, smoothness level x^δ, the bound N^k with $N = \lfloor 5x \rfloor_+$ for the values, and the explicit choice $\delta = \eta = \varepsilon/8$.
- *The value count.* The source uses the bound $(1 + k \log X / \log 2)^{X^\delta}$. The formalization uses the explicit bound $(\ell_2(N^k) + 1)^{\lfloor x^\delta \rfloor_+}$. The statement "$= X^{o(k)}$" is replaced by the explicit inequalities of Lemmas 2.9, 2.11, 2.12 and 2.16.
- *Unboundedness of n.* The source deduces this from the growth of $g(n)$ and finiteness of fibers. The formalization shows directly that $n \geq p_0 - 1 > x$ for some prime p_0 in the chosen product (Lemma 2.18). Finiteness of fibers (Lemma 2.1) is used only to know that $g(n)$ is a genuine count (Lemma 2.2).
- *Lemma 1 of the source.* The quantitative bound $g(n) \ll_\eta n^{1+\eta}$ is neither formalized nor needed.
- *The hypothesis.* Hypothesis (H), Theorem 1.2 of the source paper, enters as an explicit assumption, as in the problem statement, and is not proved.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.4 is `openaimath_011_totientfibersfromsmo`

Lemma	Lean declaration	Status
Lemma 2.1	`totient_fiber_finite`	verified
Lemma 2.2	`card_le_totientFiberCard`	verified
Lemma 2.3	`totient_prod_primes`	verified
Lemma 2.4	`prod_primes_inj`	verified
Lemma 2.5	`exists_large_fiber`	verified
Lemma 2.6	`smooth_count`	verified
Lemma 2.7	`prod_smooth_bounded`	verified
Lemma 2.8	`choose_ge_pow_div`	verified
Lemma 2.9	`asymp_log_ineq_growth_logbound`	verified
Lemma 2.10	`asymp_log_ineq_growth_eventually`	verified
Lemma 2.11	`asymp_log_ineq_growth`	verified
Lemma 2.12	`asymp_log_ineq_pointwise`	verified
Lemma 2.13	`asymp_log_ineq`	verified
Lemma 2.14	`asymp_main_lhs_log`	verified
Lemma 2.15	`asymp_main_rhs_log`	verified
Lemma 2.16	`asymp_main`	verified
Lemma 2.17	`exists_prime_finset`	verified
Lemma 2.18	`good_n_from_finset`	verified

Lemma	Lean declaration	Status
Lemma 2.19	`exists_good_n_large`	verified
Lemma 2.20	`infinite_of_lt_one`	verified